

An Introduction to Agent-Based Modeling

Unit 5: Components of an Agent-Based Model

Bill Rand

Assistant Professor of Business Management
Poole College of Management
North Carolina State University

So What Makes Up An ABM?

Agents

Environment

Interaction



Basic Agent Properties

WHO

COLOR

HEADING

XCOR and YCOR

SHAPE

LABEL

LABEL-COLOR

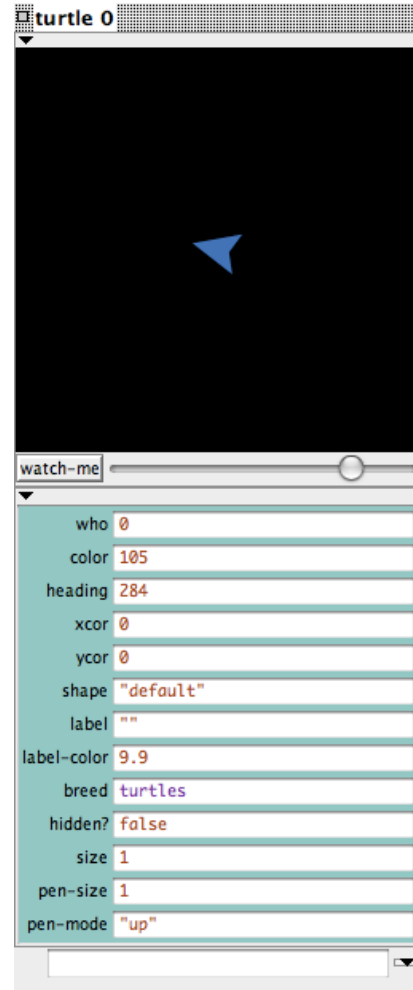
BREED

HIDDEN?

SIZE

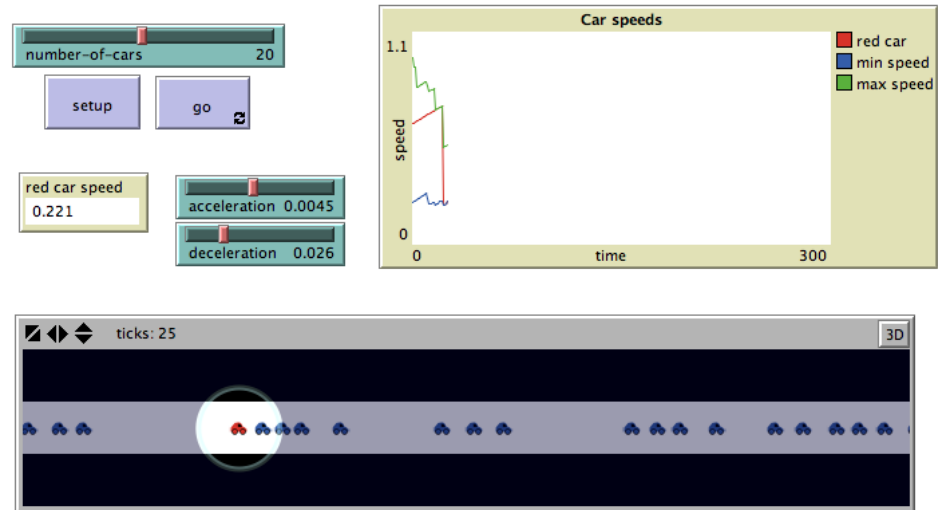
PEN-SIZE

PEN-MODE



Author-Defined Properties

- In Traffic Basic
 - speed-limit
 - speed
 - speed-min
- Types of values
 - Fixed (10)
 - Distributions (random-normal 10 1)
 - Variable (acceleration)



Agent Actions

Standard Actions

FORWARD

RIGHT

LEFT

HATCH

DIE

MOVE-TO

User-Defined Actions

speed-up-car

slow-down-car

Collections of Agents

- Built-In Collections
 - *patches, turtles, links*
 - *turtles-here, in-link-neighbors*
- Agent Breeds
 - *influentials, imitators*
 - can have their own properties
- Agentsets
 - *with*
 - *turtles-on*

Agentsets and Lists

- Lists can hold any type of item, but agentsets can only hold agents
- *no-turtles*
- We can convert from one to the other, primarily using ?

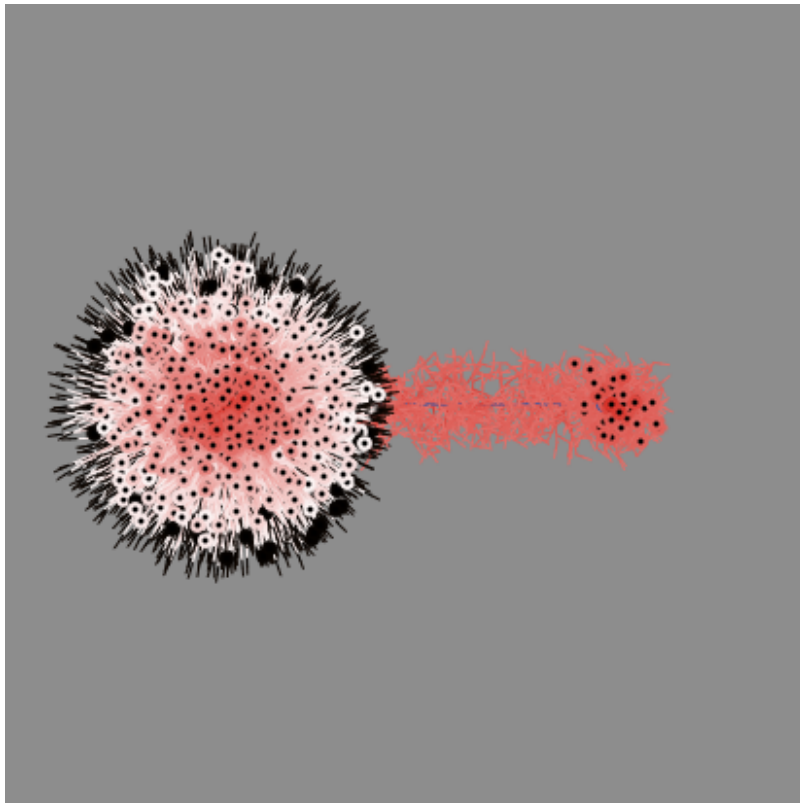
```
foreach a-list [  
    ask ? [ do something ]  
]
```

Agentsets and Computation

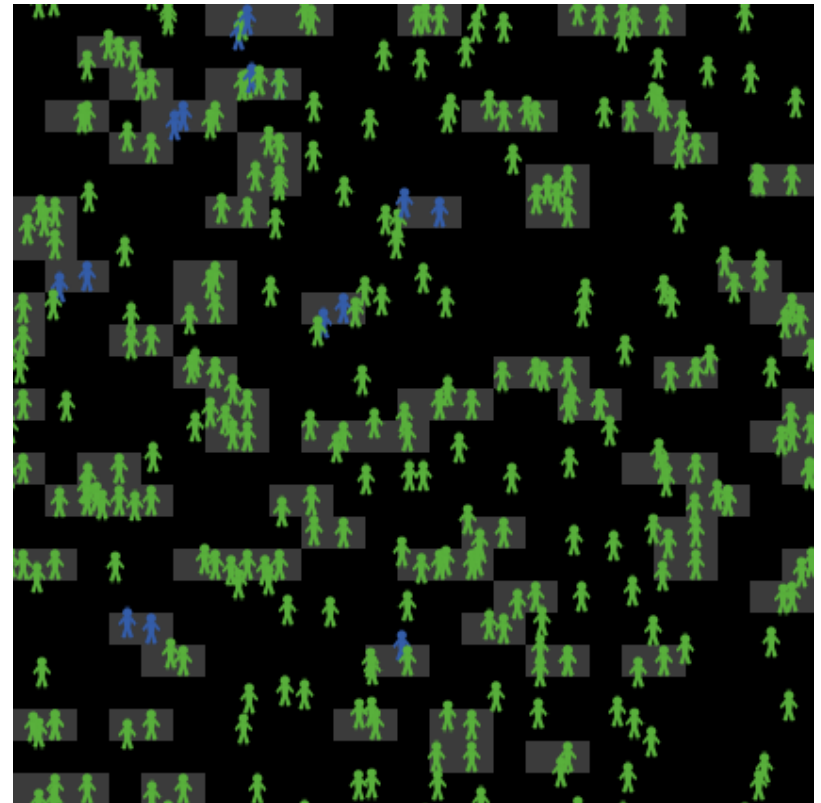
- It is important to realize that when an agentset is created it remains static until it is created again or updated
- Agentset Ordering
- Agentset Efficiency

The Level of an Agent

Tumor Model



AIDS Model



Meta-Agent

- An agent composed of other agents
- Turtles all the way down!
- *tie* (for a spatial example)

Proto-Agent

- An agent that is not fully realized
- Often built as regular agents

Agent Cognition

- *Agent cognition* is the process by which agents examine their own properties and the world around them, and then make a decision about what actions to take.
- Before constructing your model, you should consider what level of cognition the agents will have?
- The more complex the cognition, the more computational effort may be required, but potentially the more realistic the model.

Types of Agent Cognition

(Russell and Norvig, 1995)

- Reflexive
- Utility-Based
- Goal-Based
- Adaptive

Reflexive Agents

- Simple rules where agents react to what is around them
- Often represented by if-then rules
- Car in Traffic Basic

Utility-Based

- Agents are attempting to maximize some utility function
- Often requires agents to try different actions and then observe the outcome on the utility function
- Traffic Basic Utility

Goal-Based

- Agents are attempting to achieve some goal
- As opposed to utility, there is some criteria that establishes when the agent has achieved its goal
- Traffic Grid Goal

Adaptive Agent

- Agents can change not only their decisions but also their strategies
- The action that an agent takes given the same environment may be different over time based on their past experience
- Traffic Basic Adaptive

Environments

- The second main component
- A number of ways to represent
 - patches - the environment is composed of a number of spaces
 - uniform - one large agent with a uniform set of properties
 - external - could be implemented outside of the ABM environment

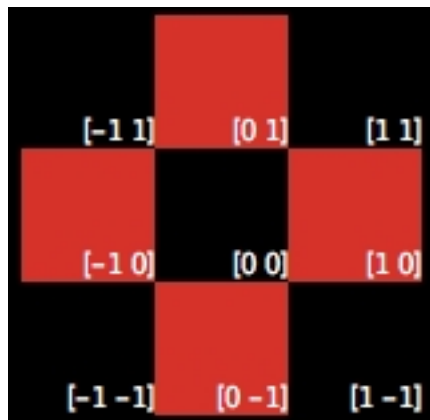
Four of the Most Popular Types of Environments

- Spatial Environments
- Network-Based Environments
- 3D Worlds
- GIS

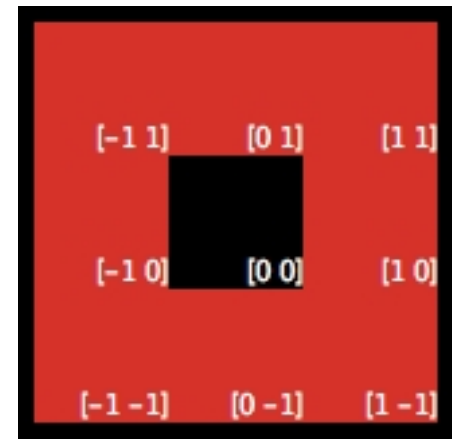
Spatial Environments

- Discrete Spaces
 - Lattice Graphs (mesh graphs or grid graphs)
 - Square Lattice

Von Neumann

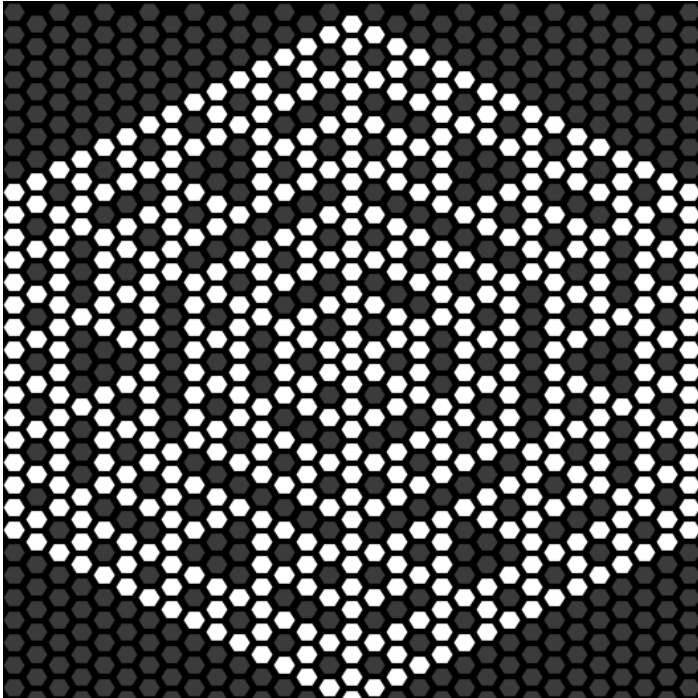


Moore

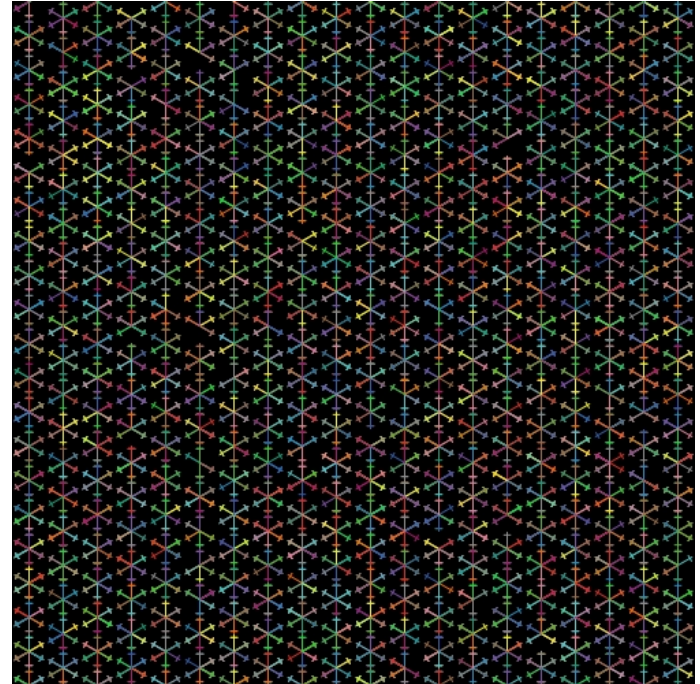


Hex Grids

Hex Cells

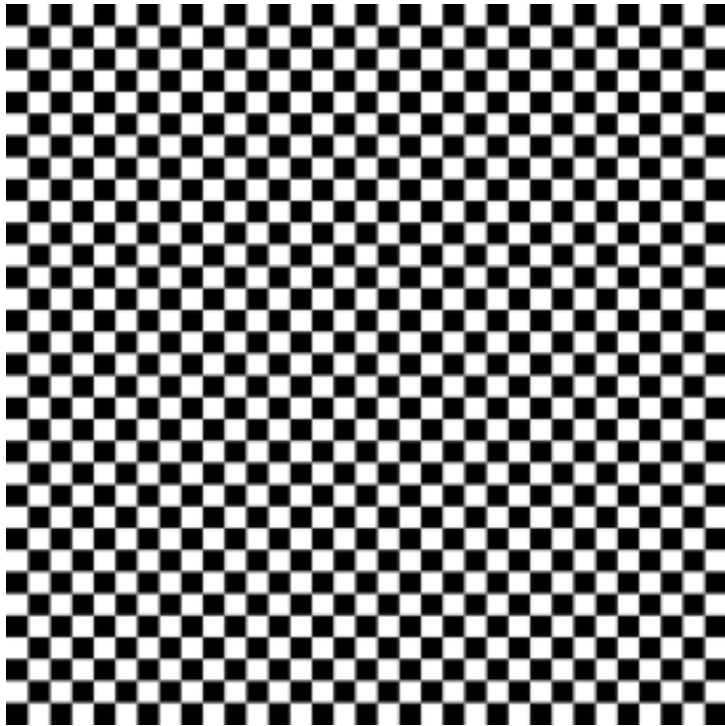


Hex Turtles



Continuous Space

- NetLogo is a continuous space with a discrete laid on top of it



```
ask patches with  
  [ (pxcor + pycor) mod 2 = 1 ]  
  [  
    set pcolor white  
  ]
```

Boundary Conditions

- Toroidal Topology

Wrapping

Boids Model

- Bounded Topology

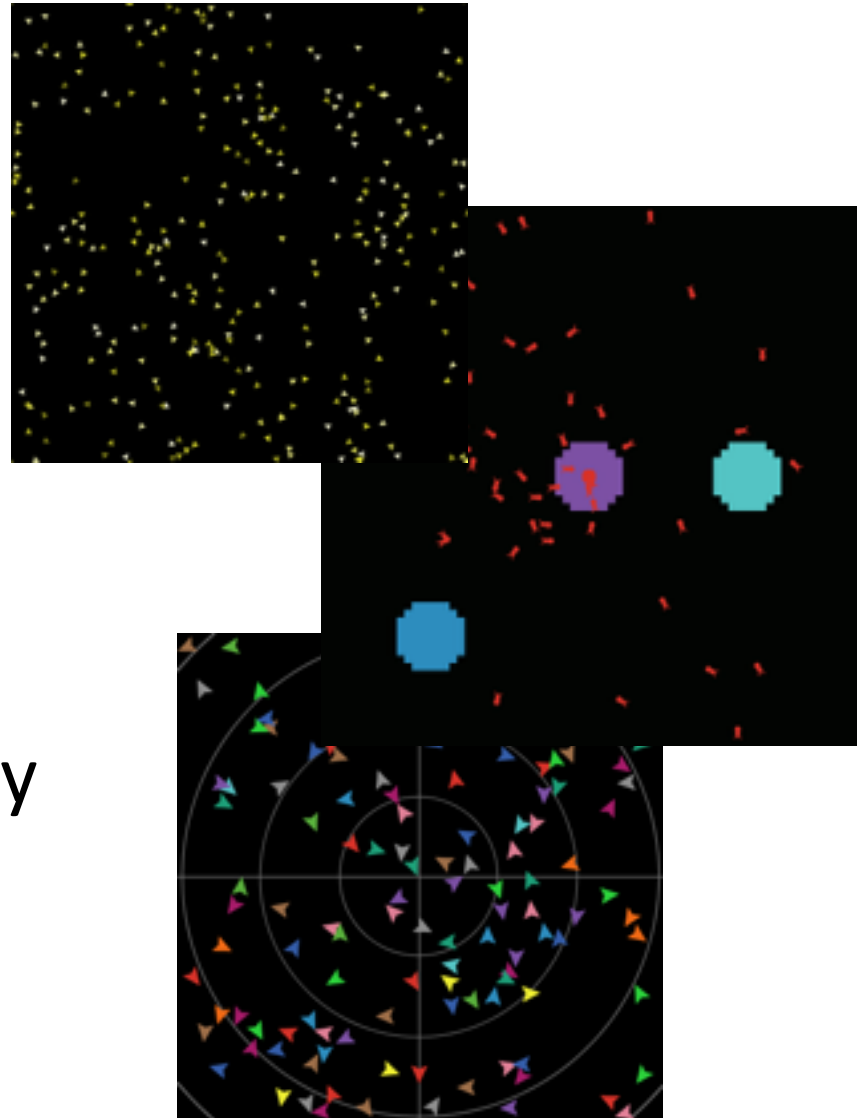
Mazes

Ants Model

- Infinite Plane Topology

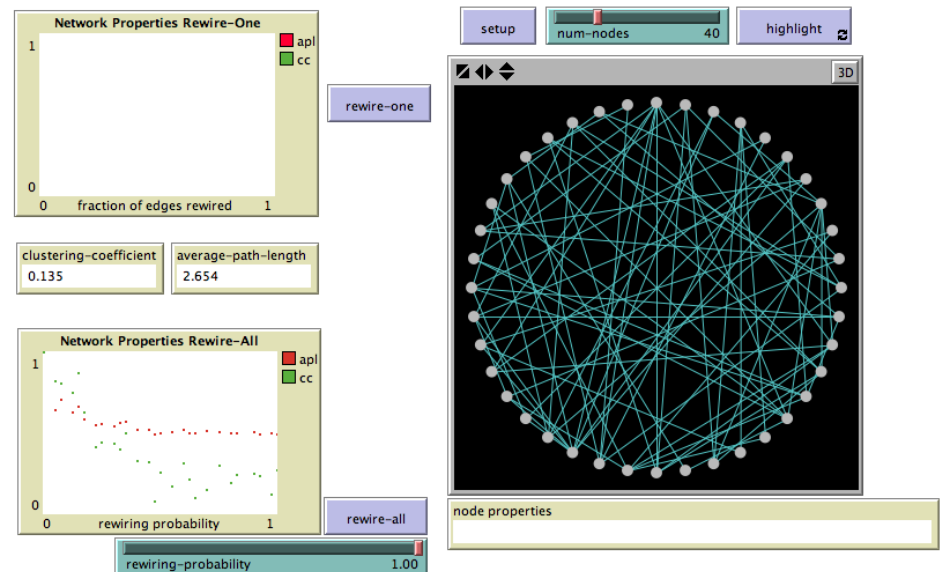
Track Agents Anywhere

Random Walk 360 model

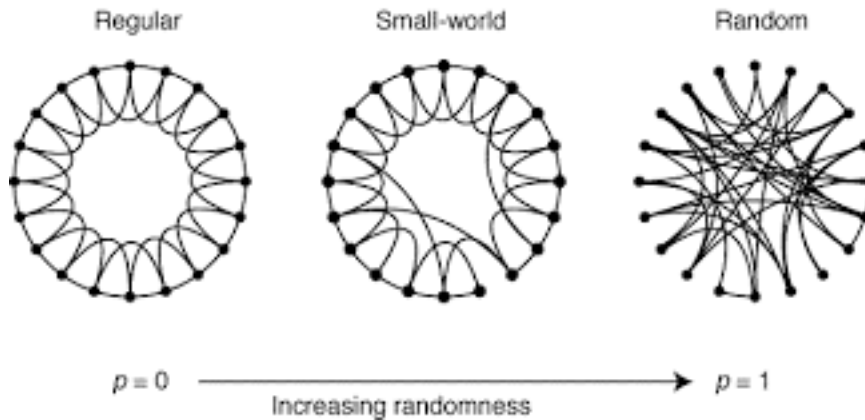


Network Environments

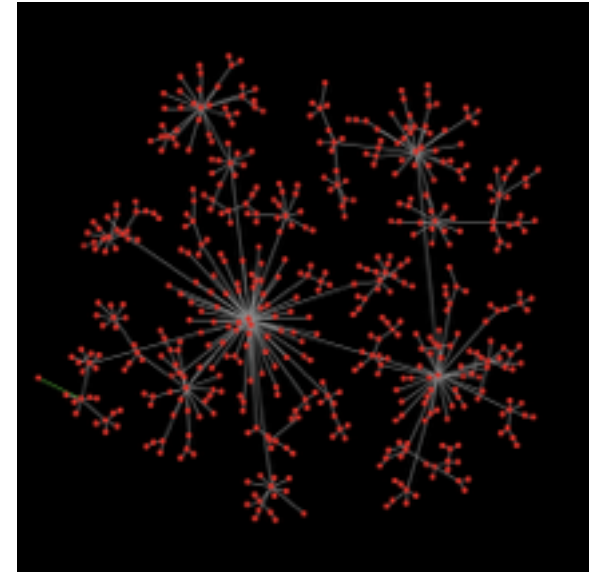
- Different Types of Networks
 - Grid Environments are Regular Networks
 - Scale-Free / Preferential Attachment
 - Random
 - Small World
 - Real World Data



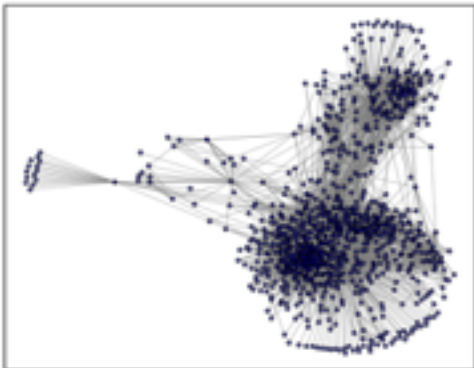
Visualizations of Networks



Watts and Strogatz, 1997



Preferential Attachment Model
Barabasi and Albert, 1999

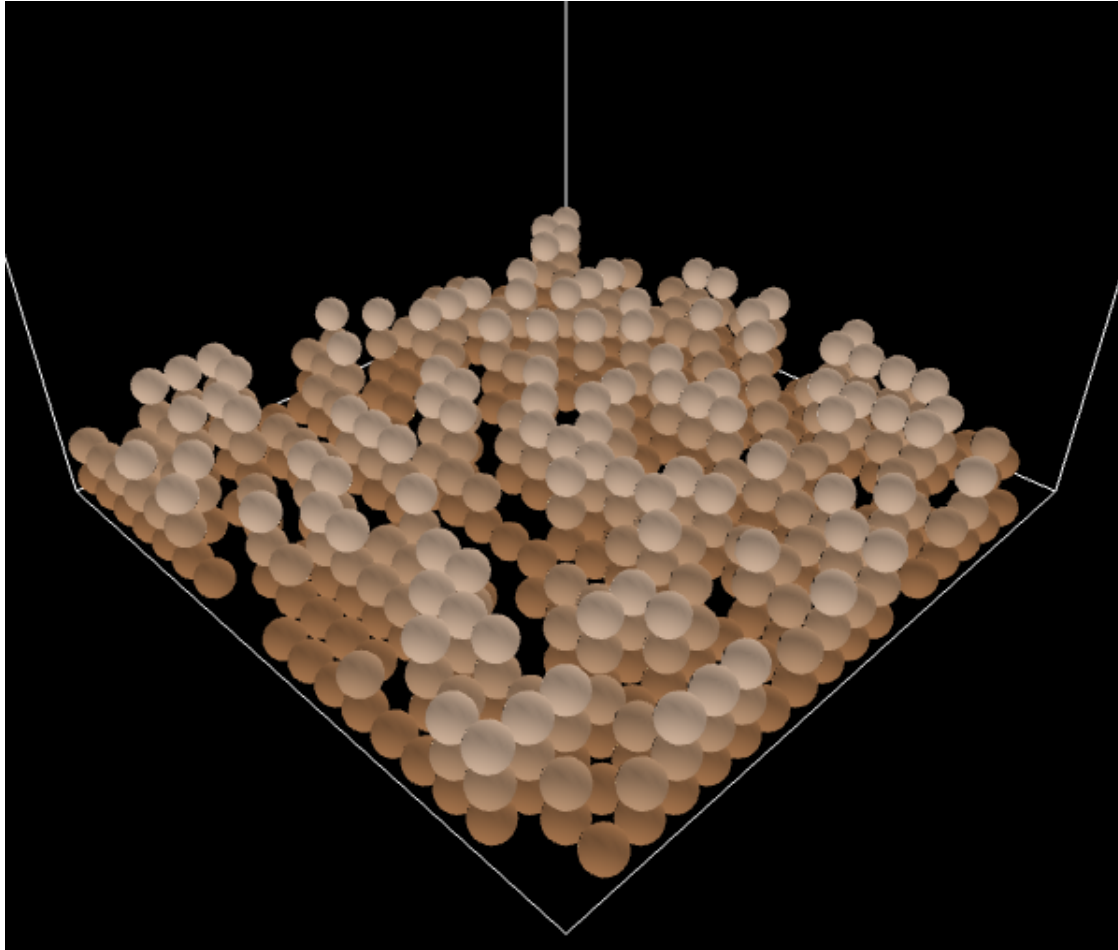


Twitter
Stonedahl et al., 2010

Different Types of Measures for Networks

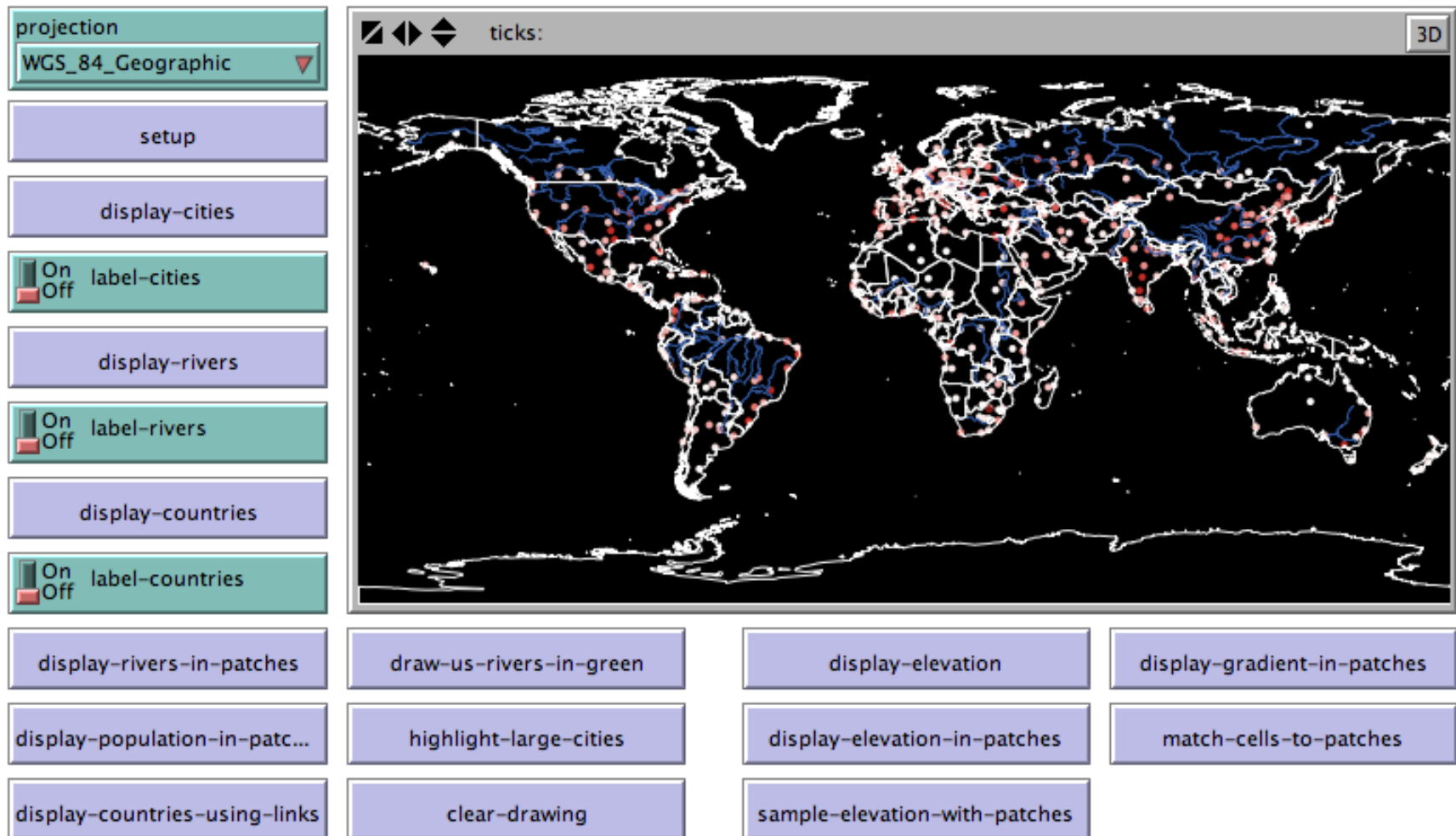
- Degree Distribution
 - How many friends should we expect a person to have?
- Average Clustering Coefficient
 - On average how many friends of my friends are my friends?
 - Out of all possible triangle connections between my friends how many exist?
- Average Path Length
 - How many friendship connections is it from any person in the network to any other person?
- Often real-world networks:
 - have surprisingly low average path lengths for a high clustering coefficient
 - tend to be power-law scaled when it comes to degree distributions

3D Environments



3D Sandpile

GIS Environments



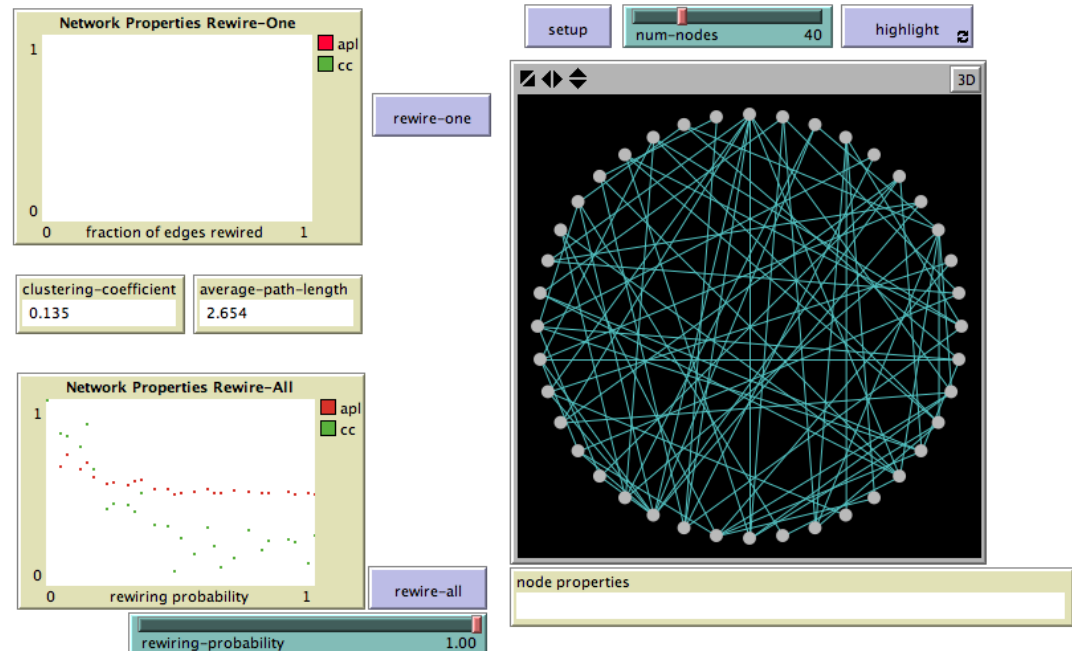
GIS General Examples

Interactions (the third main element)

- Agent-Self Interactions
 - Agents can interact with themselves
 - Checking to see if a sheep has enough energy to reproduce
- Environment-Self Interactions
 - Patches can interact with their own state variables
 - Regrowing grass within a patch
- Agent-Agent Interactions
 - Agents can interact with other agents
 - Wolves eating sheep
- Environment-Environment Interactions
 - Parts of the environment can interact with each other
 - Spatial Diffusion
- Agent-Environment Interaction
 - The agent can interact with the environment
 - Sheep eating grass

Observer / User Interface

- Design the interface well
- Make sure buttons and sliders are placed where it makes sense to place them



Creating Good Visualizations

Kornhauser et al., 2007

- Simplify the Visualization
 - Remove unwanted clutter
- Explain the Components
 - Make sure it is obvious why each element is there
- Emphasize the main point
 - All models tell a story, make sure the story is obvious

Batch vs. Interactive Interaction

- Interactive
 - Normal way of using NetLogo
 - via immediate use of the Graphical User Interface
- Batch
 - Running many models at once can be done either:
 - via the GUI ala BehaviorSpace
 - via the command line via Headless Running

Schedule

- The *schedule* is a description of the order of events in which the model operates
- SETUP and GO
- Synchronous vs. Asynchronous Updating
 - Asynchronous: Traffic Basic, Wolf Sheep, Ants, Segregation
 - Synchronous: Cellular Automata, Ethnocentrism
- Sequential vs. Parallel Actions
 - Sequential: Agents take actions in turns
 - Parallel: Agents operate simultaneously (Termites)

Unit 5 Overview

- Agents
- Agentsets
- Agent Granularity
- Agent Cognition
- Meta-Agents and Proto-Agents
- Spatial and Network Environments
- 3D and GIS Environments
- Interactions
- Interface
- Schedule
- Unit 5 Slides
- Course Feedback
- Unit 5 Test